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The Editor-in-Chief
Computer Methods and Programs in Biomedicine

Dear Editor,

I am pleased to submit the original research article **"Siamese metric learning for generalised few-shot recognition of rare viruses in transmission electron microscopy images"** for consideration in *Computer Methods and Programs in Biomedicine*.

Why this journal. The public TEM virus dataset and its benchmark were introduced in this journal by Matuszewski and Sintorn (2021). This work is a direct methodological follow-up on that dataset and its official split. It develops and evaluates a computing method for a diagnostic imaging task, and releases the software, protocol and per-patch predictions for reuse. This matches the journal's focus on computing methods and software for biomedical research and practice.

What is known. The benchmark covers the 14 most frequent viruses. Subsequent studies have reported higher 14-class accuracy, but several used protocols other than the official micrograph-level split. The two studies that used all 22 classes trained a closed-set classifier on every class. They reported neither per-class results for the eight rare viruses nor results on the official split. Yet rare viruses, with only 5–25 training micrographs each, are the realistic diagnostic case: an emerging virus will always have few reference images.

What this study adds.

- **A first protocol for the rare viruses.** This is the first generalised few-shot evaluation of all 22 classes on the official split. The eight rare viruses are never used for training, every hyper-parameter is chosen on validation, and a pixel-level audit confirms there is no train/test leakage.
- **A fair comparison on the benchmark.** A Siamese supervised-contrastive encoder matches a faithful re-implementation of the benchmark classifier on the 14 classes: $94.79 \pm 0.21\,\%$ versus $93.91 \pm 0.54\,\%$ accuracy over three seeds each, with no significant seed-pair difference. Both exceed the published $93.1\,\%$.
- **A one-shot advantage.** With a single example per rare virus, the Siamese features recognise rare viruses better than cross-entropy features ($37.3 \pm 2.3\,\%$ vs $30.8 \pm 2.4\,\%$ rare-class accuracy). The difference is significant in all seed pairs.
- **A diagnosis and a simple fix for the rare-class failures.** The confused rare classes, including Dengue versus Sapovirus, are nearly linearly separable in the learned features. A validation-selected logistic regression on frozen features therefore raises 22-class macro-F1 from 0.786 ± 0.008 to 0.849 ± 0.009, significantly on every checkpoint, without retraining.
- **Transparent reporting.** Negative and neutral results are reported in full, including where the Siamese approach gives no advantage. Statistics use micrograph-level cluster bootstrap and paired tests.

Declarations. I confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. I am the sole author and approve the submission. I have no competing interests to declare, and this research received no specific funding. The dataset is publicly available under a CC BY 4.0 licence, and no human or animal subjects were involved. As required by the journal's policy, the use of generative AI tools in preparing the work is declared in the manuscript.

Thank you for considering this manuscript. I look forward to hearing from you.

Sincerely,

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